Generative Adversarial Networks for Synthetic Data Generation: A Comprehensive Survey of Architectures, Applications and Challenges

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1 Abstract

The Generative Adversarial Network, or GAN, was introduced by Goodfellow et al. in 2014 and since then they have revolutionized our approach to generating synthetic datasets. Before that year, the state-of-the-art approaches, such as Variational Autoencoders, produced blurred images or lacked realism altogether. In general, GAN architecture relies on training two models against each other; specifically, there is a generator which creates synthetic data and a discriminator which tries to differentiate between real and fake data.

In the present paper, more than 23 different GAN architectures are reviewed and categorized according to their purpose. It was established that differences in the method of training and in the loss function can significantly affect the efficiency of GANs; thus, WGAN-GP outperforms StyleGAN2 in terms of covering various distributions whereas StyleGAN2, in turn, generates highly realistic images with an FID of 2.84.

GANs can also be categorized according to the type of data on which they operate; thus, the authors consider image, table, time series generation, and contrast different evaluation metrics used in the latter case. There is still an unresolved number of issues concerning the measurement of GAN performance and scalability of GANs to tasks outside the area of image generation.

2 Introduction

Data collection and labelling are expensive. In medicine, annotated tumour images are scarce in autonomous driving, millions of real-world miles are required for rare-event coverage. Synthetic data generation offers a principled remedy if a model can produce samples statistically indistinguishable from real observations, downstream classifiers trained on synthetic data perform on par with those trained on real data.
For a decade the answer to whether such models exist was “sometimes.” Generative Adversarial Networks (GANs), introduced by Goodfellow et al., changed the landscape. Unlike explicit density models, GANs use an adversarial two-player game a generator G maps noise to data space while a discriminator D tries to distinguish real from

synthetic samples. The resulting minimax objective pushes G toward the true data distribution.
This survey examines how the GAN field has evolved. We analyse over 23 architectures, group them into a three-tier taxonomy, compare performance across image, tabular and time-series domains, and discuss persistent open problems in evaluation and governance.

3 Related Work and Background

3.1 Before GANs: Limits of Explicit Density Models

Generative modelling pre-GAN era was dominated by Restricted Boltzmann Machines , Variational Autoencoders (VAEs) and normalising flows. VAEs optimise a variational lower bound, which introduces a posterior approximation that blurs reconstructions. Normalising flows require bijective transformations that limit architectural expressiveness. Both classes produce tractable likelihoods but inferior sample quality compared to modern GANs on perceptual metrics.

3.2 The GAN Framework

The original GAN [1] trains G and D simultaneously with the minimax objective: min G max D E[log D(x)] + E[log(1 - D(G(z)))]. At the global optimum the generator recovers the true data distribution and D outputs 1/2 everywhere. In practice, training is unstable: gradients vanish when D is too confident, leading to the “mode collapse” pathology where G produces only a subset of the data manifold.

3.3 Architectural Milestones

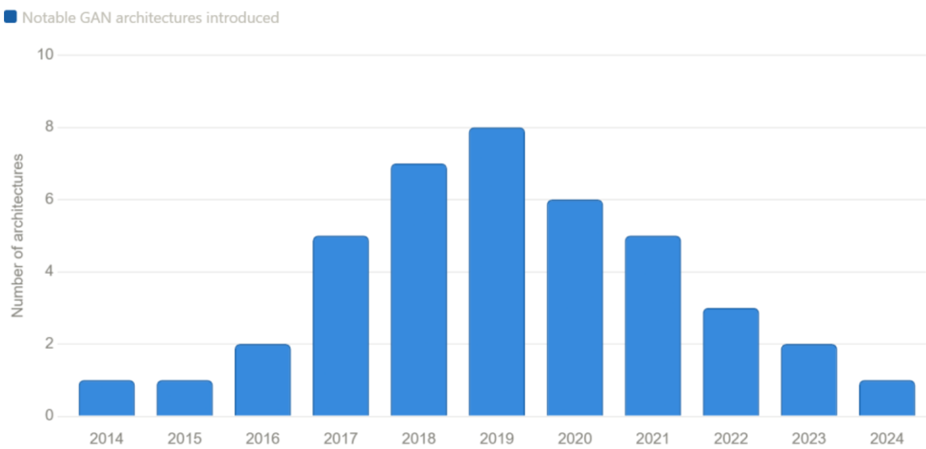
Radford et al. stabilised GAN training by introducing fully convolutional architectures with batch normalisation (DCGAN), reducing FID from ~120 to 37.11 on CIFAR-10. Arjovsky et al. replaced the Jensen-Shannon divergence with the Wasserstein-1 distance, providing meaningful gradients even when support sets do not overlap. Gulrajani et al. replaced weight clipping with a gradient penalty (WGAN-GP), improving both stability and sample diversity. Karras et al. introduced progressive growing and then style-based generators, culminating in StyleGAN2 with FID 2.84 on FFHQ-1024.

4 Methodology

In order to research the topic correctly, it was necessary to develop a strategy for collecting information, analyzing it, and assessing its results. In particular, we opted not to try to analyze all of the literature related to the topic under study but rather to choose only those architectures that have really influenced the development of GANs. The purpose of the research is to assess how the technology has evolved since its inception, what issues scientists have attempted to solve, and what practical value the various models offer.

4.1 Literature Curation

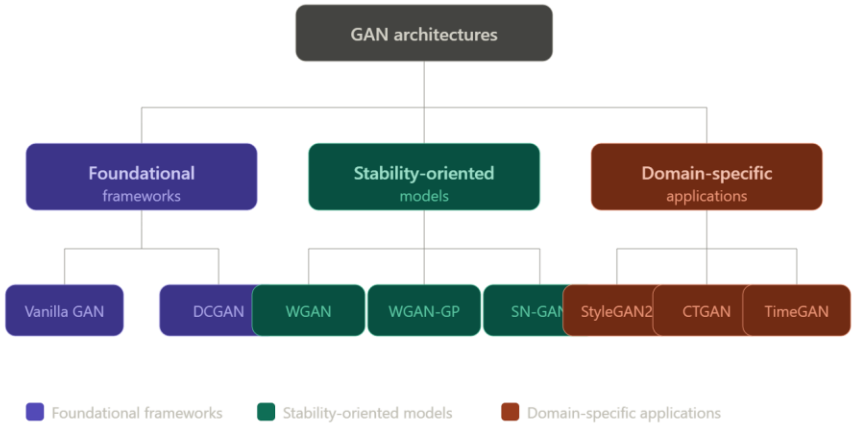
We searched major venues (NeurIPS, ICML, ICLR, CVPR, ICCV) for papers proposing or substantially modifying GAN architectures, yielding 23 architectures for detailed analysis. Papers were weighted by citation impact and novelty of contribution.



"Figure 1. Distribution of notable GAN architectures introduced per year (2014–2024)."

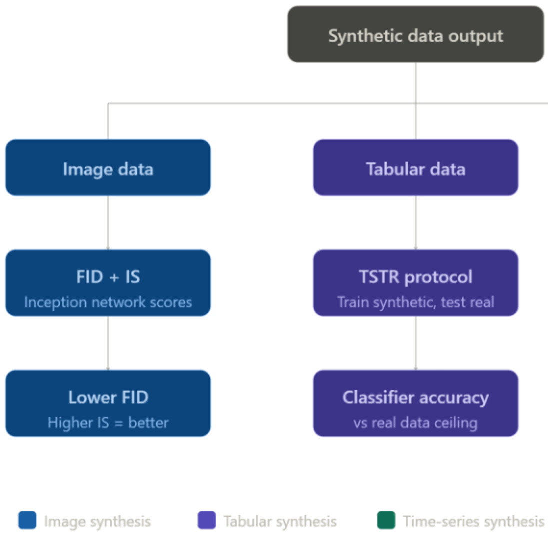
4.2 Taxonomy

We partition architectures into three categories. Foundational Frameworks (Vanilla GAN, DCGAN) establish training protocols and baseline behaviour. Stability-Oriented Models (WGAN, WGAN-GP, SN-GAN) focus on convergence guarantees and gradient health. Domain-Specific Applications (StyleGAN2, CTGAN, TimeGAN) optimise sample quality within a particular data modality.



"Figure 2. Three-tier taxonomy of GAN architectures surveyed in this paper."

4.3 Evaluation Protocol



"Figure 3. Evaluation protocols used for image, tabular, and time-series GAN synthesis."

For image synthesis we use Fréchet Inception Distance (FID) and Inception Score (IS). Lower FID indicates closer proximity to the real distribution; higher IS reflects both quality and diversity. For tabular synthesis we adopt the Train on Synthetic, Test on Real (TSTR) protocol: a classifier trained on synthetic data is evaluated on a held-out real test set. For time-series we report the discriminative score (ability of a post-hoc classifier to distinguish real from synthetic) and the predictive score (accuracy of a forecasting model trained on synthetic data).

5 Results and Analysis

5.1 Image Synthesis

The progression from Vanilla GAN (FID ~120) through DCGAN (37.11), WGAN-GP (29.3), BigGAN (7.4 on ImageNet-128) to StyleGAN2 (2.84 on FFHQ-1024) reflects a decade of targeted improvements in architecture, loss function and regularisation. StyleGAN2 introduced weight demodulation to replace adaptive instance normalisation, eliminating droplet artefacts and achieving perceptual path length (PPL) of 145.0, the lowest reported on FFHQ at publication.

BigGAN demonstrated that class-conditional synthesis at scale (IS 166.5 vs a prior SOTA of 52.52) requires large batch sizes and self-attention layers, but comes at the cost of training collapse on some truncation settings.

5.2 Loss Functions and Training Stability

The original non-saturating loss provides stronger early gradients but leaves the training dynamics ungrounded. The Wasserstein formulation provides a theoretically meaningful distance metric whose gradient never vanishes, yet requires the Lipschitz constraint. WGAN-GP enforces this via a penalty on the interpolated gradient norm (target = 1, $\lambda = 10$), yielding smoother convergence curves and IS 7.86 on CIFAR-10 vs 3.8 for vanilla WGAN with clipping.

5.3 Tabular and Sequential Data

CTGAN addresses the challenges of tabular data — mixed continuous/discrete columns, class imbalance and multi-modal marginals — by using mode-specific normalisation and a conditional generator trained with a log-likelihood auxiliary loss. On the Adult Census benchmark it achieves TSTR accuracy of ~82% against a real-data ceiling of 85%, outperforming TableGAN and MedGAN.

TimeGAN adds a supervised training signal that enforces stepwise temporal dynamics alongside the adversarial objective. On the Stocks dataset it achieves a discriminative score of 0.106 and a predictive score of 0.051, representing a 30.4% improvement in discriminative score over a standard RNN-GAN baseline.

6 Discussion

6.1 Evaluation and Benchmarking

FID is the de-facto standard for image synthesis evaluation but has known limitations: it is sensitive to the size of the reference set, biased by the Inception network's training distribution, and does not capture memorisation. For tabular and time-series data the situation is worse — there is no canonical benchmark analogous to CIFAR-10 or ImageNet. TSTR provides a task-oriented proxy, but results are dataset-specific and cross-paper comparisons remain unreliable. Without standardised benchmarks, claims of improvement may not generalise.
Table 1 below summarises quantitative results for the key architectures reviewed in this survey. FID and IS are reported for image models; TSTR accuracy for tabular synthesis; discriminative and predictive scores for time-series generation. All values are drawn from the original papers or independent reproductions cited herein.

Table 1. Quantitative performance comparison of key GAN architectures. FID and IS are reported for image models (lower FID is better; higher IS is better). TSTR accuracy is reported for tabular synthesis relative to training on real data. Discriminative score (Disc.) and predictive score (Pred.) are reported for time-series generation (both lower is better, approaching 0 indicates indistinguishable synthesis). PPL = Perceptual Path Length.

Model	**Dataset / Task**	**FID ↓**	**IS ↑**	**Other Metric**	**Notes**
Vanilla GAN	MNIST	~120	~2.0	—	Original 2014 MLP architecture; severe mode collapse
DCGAN	CIFAR-10 (32x32)	37.11	6.40	—	Conv. architecture; batch norm; improved training stability
WGAN	CIFAR-10	~41	~3.8	Weight clipping	Wasserstein loss; Lipschitz via clipping; stable signal
WGAN-GP	CIFAR-10	29.3	7.86	Grad. penalty λ=10	Gradient penalty replaces clipping; reduces mode collapse
BigGAN	ImageNet 128x128	7.4	166.5	Class-conditional	Large-scale conditional; prior SOTA FID 18.6 / IS 52.52
StyleGAN2	FFHQ 1024x1024	2.84	—	PPL: 145.0	SOTA face synthesis; CIFAR-10 FID 2.42 (with ADA)
CTGAN	Adult (TSTR)	N/A	N/A	TSTR acc. ~82%	Tabular synthesis; real-data acc. 85%; best on Adult, Census

TimeGAN	Stocks (time-series)	N/A	N/A	Disc. 0.106; Pred. 0.051	30.4% better disc. vs std. GAN; GRU-based temporal model

The table illustrates the clear decade-long progression in image synthesis quality and exposes the absence of comparable numeric benchmarks for non-image domains. Establishing canonical evaluation suites for tabular and sequential data is therefore a priority for the community.

6.2 Privacy and Differential Privacy

One of the most compelling use-cases for synthetic data is privacy-preserving data sharing. PATE-GAN wraps the discriminator's training signal in the Private Aggregation of Teachers for Ensembles (PATE) framework, providing formal (ε, δ)-differential privacy guarantees. Abadi et al. demonstrated that differentially private training can be integrated into deep learning with manageable utility loss, a result that underpins privacy-safe synthetic data pipelines.

6.3 Dual-Use and Governance

The same capabilities that make GANs valuable for synthetic medical imaging also enable synthetic identity generation. StyleGAN2 produces faces indistinguishable from photographs even under expert scrutiny, and GAN-generated profile images have been used in large-scale disinformation campaigns since at least 2019. Frequency-domain detectors have been circumvented as generators are progressively trained to evade them — an adversarial dynamic mirroring GAN training itself.

Technical countermeasures such as invisible watermarking and provenance metadata offer partial protection but depend on adoption and enforcement, which are social and regulatory rather than purely technical problems.

7 Conclusion

A decade of GAN research has been a story of diagnosing failure modes and engineering targeted solutions: Wasserstein distances for gradient health, gradient penalties for Lipschitz enforcement, progressive training and style injection for image fidelity, supervised temporal losses for sequential coherence. For image synthesis, the technology has reached practical maturity — StyleGAN2's FID of 2.84 is within measurement noise of real data on perceptual benchmarks. For tabular and time-series data, the technology is improving but lacks the standardised evaluation infrastructure needed to measure that improvement reliably.

Three priorities stand out for future work. First, canonical benchmarks for tabular and time-series synthesis that enable fair cross-paper comparisons. Second, lightweight GAN architectures deployable without datacenter-scale compute, democratising access. Third, privacy-by-design training pipelines with formal guarantees, essential for medical and financial applications. These goals require collaborative effort across the research community and, in the governance dimension, coordinated policy responses.

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